

TankLevel_LD

Tank level simulation with manual valve control and alarms (Ladder Diagram)

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Tags

inputs:

- name: FillButton
 dataType: BOOL
- name: DrainButton
 dataType: BOOL

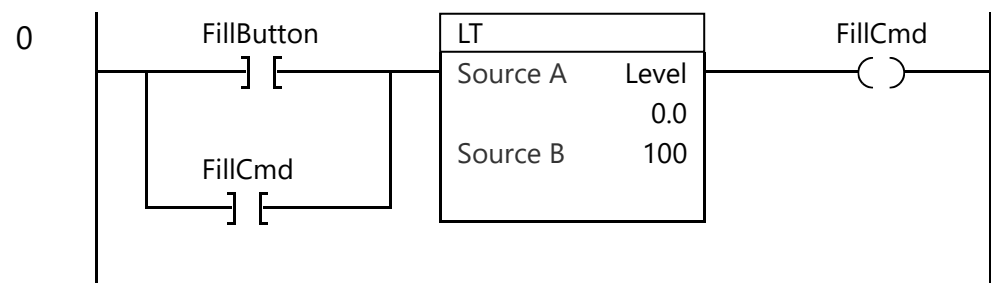
outputs:

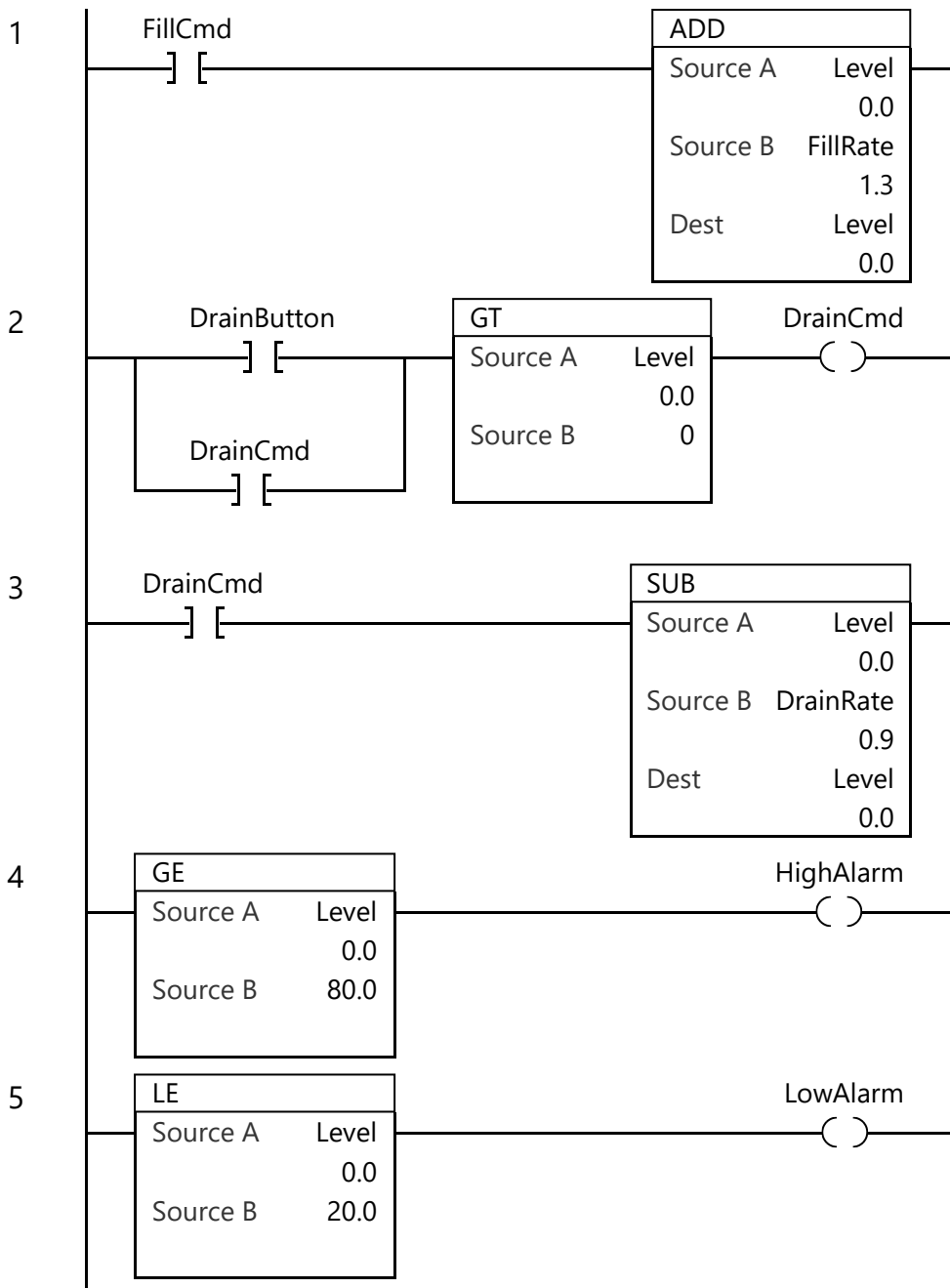
- name: Level
 description: Current tank level (0-100%)
 dataType: REAL
- name: HighAlarm
 description: High level alarm (above 80%)
 dataType: BOOL
- name: LowAlarm
 description: Low level alarm (below 20%)
 dataType: BOOL

locals:

- name: FillCmd
 description: Command to open fill valve
 dataType: BOOL
- name: DrainCmd
 description: Command to open drain valve
 dataType: BOOL
- name: FillRate
 description: Rate at which tank fills per scan
 dataType: REAL
 defaultValue: 1.3
- name: DrainRate
 description: Rate at which tank drains per scan
 dataType: REAL
 defaultValue: 0.9

Logic





Tests

- name: increases level when fill is active
 - steps:
 - in:
 - FillCmd: 1
 - advance:
 - scans: 1
 - expect:
 - Level: 51.3
- name: decreases level when drain is active
 - steps:
 - in:
 - DrainCmd: 1
 - advance:
 - scans: 1
 - expect:

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    Level: 49.1
- name: triggers high alarm above 80
  steps:
    - in:
        FillCmd: 1
      advance:
        scans: 24
      expect:
        HighAlarm: 1
- name: triggers low alarm below 20
  steps:
    - in:
        DrainCmd: 1
      advance:
        scans: 34
      expect:
        LowAlarm: 1
- name: clamps level at 100
  steps:
    - in:
        FillCmd: 1
      advance:
        scans: 40
      expect:
        Level: 100
- name: clamps level at 0
  steps:
    - in:
        DrainCmd: 1
      advance:
        scans: 57
      expect:
        Level: 0
```